RHCSA Study Notes — 08 August 2026

Conversation summary from approximately 12:40 PM onward

1. What I observed about the user's learning approach

- The user usually knows the command or basic solution and wants to understand the meaning behind each part rather than memorize commands blindly.
- A recurring difficulty is decoding RHCSA question wording into the correct Linux concept. The user explicitly noticed that some questions are technically easy but the wording can be confusing.
- The user tests concepts directly in the RHEL lab and brings terminal output back for verification.
- The user pays attention to small command details such as + / - in LVM, % in sudoers, directory versus file permissions, and whether an option is actually necessary.
- The user prefers simple, exam-oriented explanations and mental rules that can be remembered during the RHCSA exam.

2. systemd — checking whether a service is enabled

Question: How to identify whether httpd is enabled from systemctl status output?

Key observation:

Loaded: loaded (...; enabled; vendor preset: disabled)

The actual state is 'enabled'. 'vendor preset: disabled' is only the vendor's default preset and does not mean the current service is disabled.

- systemctl is-enabled httpd → checks whether it starts automatically at boot.
- systemctl is-active httpd → checks whether it is currently running.
- enabled + active means the service is configured for boot and is currently running.

3. Q20 — Creating a 400 MiB swap partition

Main doubts: fstab mount-point value, partition type, mkswap, and swapon.

- A swap partition does not use /swap as a mount directory.
- Typical fstab entry: UUID=<uuid> swap swap defaults 0 0
- swap in the fstab mount-point field means swap space; it is not a normal directory.
- For a proper RHCSA solution, create the partition, set its partition type to Linux swap, run mkswap, add it to fstab, and activate it with swapon -a.

- Technically, mkswap + a correct fstab entry + swapon -a can make the partition work even if the fdisk partition type was not changed, but changing the partition type is the clean/conventional solution.
- mount -a is not the command used to activate swap; swapon -a is.

4. Q21 — Resizing an LVM logical volume

Question: Resize LV 'vo' in VG 'myvg' to approximately 300 MB, with only 270–290 MB accepted.

- -L 280M means make the final LV size 280M.
- -L +280M means add 280M to the current size.
- Example: current LV 200M + -L +300M = 500M, which would fail a 270–290M target.
- For a 200M LV, lvresize -r -L 280M /dev/myvg/vo makes the final size approximately 280M.
- -r resizes the filesystem together with the LV when supported.
- If the question says 'increase by 80M', use +80M. If it says 'make the final size 280M', use 280M.

5. Reducing an LV

Question: What is the process if an LV must be reduced by 90/100 MB?

- Check the filesystem first with lsblk -f or df -Th.
- XFS cannot be shrunk in place; ext3/ext4 can be reduced.
- For a shrink-by amount, -L -90M means subtract 90M from the current LV size.
- Example for a supported filesystem: lvreduce -r -L -100M /dev/VG1/LV1
- Normally unmount the filesystem before shrinking and verify that nothing is using the LV.
- After shrinking, mount it again and verify with lvs and df -h.

6. Troubleshooting the LV reduction test

The user tested: lvreduce -r /dev/VG1/LV1 -L -100M

The error included: '/dev/mapper/VG1-LV1 is in use' and e2fsck could not continue.

- Initially /database was mounted on the LV.
- fuser -vm /database showed 'kernel mount /database', which indicated the filesystem was mounted rather than a normal user process using it.
- After umount /database, findmnt /dev/mapper/VG1-LV1 produced no output, indicating it was no longer mounted.

- The general lesson is: for a reduction, ensure the filesystem is unmounted and check whether the block device is still in use with fuser/lsof; also check swapon --show in case the LV is being used as swap.

7. Q16 — Sudo access without a password

Question: Make all members of the elite group sudo users with no password facility.

- Preferred approach: create a separate sudoers file such as /etc/sudoers.d/elite.
- Rule for a group: %elite ALL=(ALL) NOPASSWD: ALL
- % means group; without % the name refers to a user.
- The filename does not have to match the group name. For example, /etc/sudoers.d/myrandomname can contain a rule for %arcitect.
- For a user, use: username ALL=(ALL) NOPASSWD: ALL
- Validate a sudoers file with visudo -cf /etc/sudoers.d/<filename>.

8. Understanding sudo, wheel, and password caching

- Both abhijeer and freddy were members of wheel and sudo -l showed (ALL) ALL for both.
- (ALL) ALL means they can run commands as another user, including root, with normal sudo authentication.
- (ALL) NOPASSWD: ALL would mean passwordless sudo.
- sudo -k clears cached sudo authentication. After sudo -k, the next sudo command should request the user's password.
- sudo whoami returning 'root' does not mean the login session changed to root. It means that particular command was executed with root privileges.
- whoami in the normal shell returns abhijeer/freddy; sudo whoami returns root.
- The user's terminal test confirmed that abhijeer and freddy both authenticate normally with sudo.

9. Q15 — Shell script, find, and SGID

Question: Create /bin/script.sh to find files in /usr/local larger than 3K and smaller than 5K, copy them to /root/d1, and set SGID on the directory.

Recommended script:

```
#!/bin/bash
mkdir -p /root/d1
find /usr/local -type f -size +3k -size -5k -exec cp -pf {} /root/d1/ \;
chmod g+s /root/d1
```

- -type f explicitly limits find to regular files.

- -size +3k means greater than 3 KiB; -size -5k means less than 5 KiB.
- -perm is not needed for this question because the search condition is based on size, not permissions.
- chmod g+s /root/d1 sets SGID on the directory. Recursive -R is unnecessary when only the directory itself is required.
- The user asked where -perm is useful; it is a find filter for permissions, such as finding SUID/SGID files.

10. Q14 — Protected shared directory and sticky bit

Question: Create /shared/projects; ownership should be alex; no user other than the owner can delete a file inside it.

Clean solution:

```
mkdir -p /shared/projects
chown alex /shared/projects
chmod +t /shared/projects
```

- The sticky bit is the key requirement when the question says users other than the owner must not delete files in the directory.
- Directory ownership and file ownership are different concepts.
- If root creates a file in /shared/projects, the file normally belongs to root:root. Making the directory owned by alex does not automatically make new files owned by alex.
- The wording 'File ownership should go to user alex' was recognized as ambiguous/poor wording; in context it was interpreted as the directory should be owned by alex.

11. File vs directory permissions — key RHCSA model

This was identified as an important area to memorize because permissions have different meanings on files and directories.

Permission	Regular file	Directory
r	Read file contents	List names in directory
w	Modify file contents	Create/delete/rename directory entries
x	Execute file	Enter/traverse/access directory

Critical RHCSA rules:

- Create a file in a directory → normally need wx on the parent directory.
- Create a directory → normally need wx on the parent directory.
- Delete a file → controlled mainly by the parent directory's wx permissions, not the file's own write permission.
- Rename a file → normally requires wx on the containing directory.

- cd into a directory → x permission on the directory.
- List directory names → r permission on the directory.
- Access a known file through a directory path → x permission is needed on each relevant parent directory, plus the appropriate permission on the file itself.
- Sticky bit changes who can delete/rename entries in a shared directory.

12. Core mental rules collected from today's session

- + in LVM means add; no sign means final/absolute size; - means subtract.
- swap is a special filesystem/use case, not a /swap directory.
- Directory owner does not determine the owner of newly created files.
- Sticky bit → protect deletion/renaming in shared directories.
- SGID on a directory → group inheritance for newly created files/directories.
- find -size → size filtering; find -perm → permission filtering.
- For sudoers: %group means group, username without % means user.
- sudo whoami → root means the command ran as root; it does not change the login user.
- sudo -k → clear cached sudo authentication.
- Directory permissions: r=list, w=modify entries, x=traverse.

End of today's RHCSA observation notes — 08 August 2026.